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A Short Introduction to Network Data Analysis (Part 2)

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Outline

Thanks to Dr. Eric Kolaczyk at McGill University for materials used to develop these lectures!

What is a Network?

- A system of interconnected elements (**nodes**)
- Also a (mathematical) ‘network graph’ representing such a system, or
- A visual object (‘map’).

Key Point: Choice of **nodes** and **connections** can produce very different network representations of the same ‘system’.

Ex.: Your Own Social Network. You are a **node** in your network **connected** to all of your friends, and each friend, has their own connections that you may or may not have.

Network Theories

- **Foundations:**
  - Network Visualization,
  - Characterization,
  - Partitioning.
- **Network Models:**
  - Random Graph Models,
  - Exponential Random Graph Models (ERGMS),
  - Network Topology Inference,
  - Stochastic Block Models (SBM), and
  - Latent Space Models (LSM).
- **Network Processes:** Diffusion and Epidemic Models.

Modern Network Theories and Applications

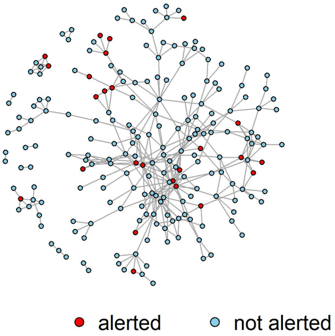
- Temporal/dynamic Network Models (e.g., TERGM).
- Multilevel and Ego Network Models.
- Deep Learning on Graphs, or Graph Neural Networks (GNNs).
- Graphons and Nonparametric Models.
- **Causal inference on Networks** (Part 3).

Applications include:

- Social networks (Facebook, LinkedIn); Biological networks (gene interactions); Epidemiology and Public Health (disease spread); Infrastructure (power grids, transportation); Finance (systemic risk), etc.

Motivational Study: TRIP

- Sociometric network-based study of injection drug users and their contacts in Athens, Greece, from 2013 to 2015.
- *Purpose:* network tracing to find recently-infected individuals and referring to HIV treatment.
- *Intervention:* Community alerts.
- *Network:* **216** nodes and **362** connections (edges).



Short Course (Part 2)

Main Goals:

1. Get familiar with Network Fundamentals.
2. Build the network object in R (similar to TRIP).
3. Characterize the network (using Degree, Clustering, Centrality measures).
4. Detect Communities.
5. Identify node attributes associated with connections in the network.
6. Run Causal Inference on Networks (in **Part 3**).

Definition of Graphs

Graphs are the mathematical objects underlying network analysis. Formally, a graph $G = (V, E)$ is a structure consisting of sets

- V of **vertices** (also commonly called **nodes**)
- E of **edges** (also commonly called **links**), where elements of E are ordered or unordered pairs $\{u, v\}$ of distinct vertices $i, j \in V$.
- Both vertices and edges can be accompanied by the sets of **attributes** X_v and X_e.

The values $N_v = |V|$ and $N_e = |E|$ are the **order** (number of vertices) and **size** (number of edges) of the graph, respectively.

Connectivity/Adjacency

- Two vertices $i, j \in V$ are said to be *adjacent* if they are joined by an edge in E.
- Two edges $e_1, e_2 \in E$ are *adjacent* if they share a common endpoint in V.
- A vertex $i \in V$ is said to be *incident* to an edge $e \in E$ if i is an endpoint of e.
- **Adjacency Matrix**, A, is a data structure for representing graphs. It is an $N_v \times N_v$ binary matrix, where

$$A_{ij} = \begin{cases} 1, & \text{if vertices } i \text{ and } j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases}$$

Node Neighborhood and Connected Components

Node Neighborhood

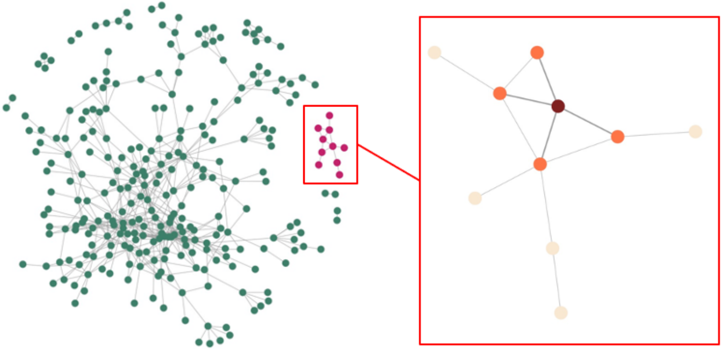
- The neighborhood of a node i is the set of nodes directly connected to it, denoted as: $|\mathcal{N}_i| = \{j : (i,j) \in E\}$.

Connected Component

- A connected component is a maximal set of nodes where every pair is connected by a path.
- *Within a component:* all nodes are reachable from each other.
- *Between components:* no paths exist.

Key idea: Neighborhood is *local*, components describe *global connectivity*.

Neighborhoods and Connected Components



The **largest connected component (LCC)** is the component with the greatest number of nodes. TRIP has **10** components of size 2-185 nodes.

Characterization of Networks

Many network questions can be framed through its structure.

- The *importance* of vertices can be captured through centrality measures.
- Communities of vertices are identified through graph partitioning.

Characterization of vertices/edges:

- Degree, density, centrality measures.

Network cohesion (connectivity/grouping)

- Clustering, assortativity, graph partitioning.

Node Degree

Node degree summarizes the connectivity of a vertex. The degree of a vertex i, d_i, is:

- the number of edges incident on i, or
- the number of neighbors of i, $|\mathcal{N}_i|$, where $\mathcal{N}_i$ is the set of participants that share a link with i.

Degree matrix D is diagonal with $D_{ii} =$ degree of node i.

The **degree sequence** of a graph G is obtained by arranging the vertex degrees in increasing order: $d_{(1)} \leq d_{(2)} \leq \cdots \leq d_{(N_v)}$.

TRIP average degree is 3.35. On average, each person in the network sharing tie with 3 to 4 other individuals.

Graph Laplacian

Adjacency Matrix, A, is an $N_v \times N_v$ binary matrix, where

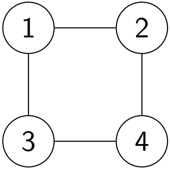
$$A_{ij} = \begin{cases} 1, & \text{if vertices } i \text{ and } j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases}$$

Degree matrix D is diagonal with $D_{ii} =$ degree of node i.
Then, L is the (unnormalized) **Graph Laplacian**:

$$L = D - A$$

- Captures connectivity and structure of the network, and
- Used in spectral clustering and for modeling diffusion processes.

4-Node Network: Adjacency and Laplacian



Adjacency matrix A

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Degree matrix D

$$D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Laplacian $L = D - A$

$$L = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

Network/Graph Density

The **density** of a graph $G = (V, E)$ measures how many edges exist relative to the maximum possible:

$$\text{Density} = \text{Density} = \frac{2|E|}{|V|(|V|-1)} \quad \text{(undirected)}$$

- **Close to 1** $\rightarrow$ very dense network (many connections).
- **Close to 0** $\rightarrow$ sparse network (few connections).

Network Density

- Captures overall connectivity level.
- Can be computed locally (e.g., component or subgraph).
- Does not reflect structure (e.g., clustering or communities).

Real-world networks are typically sparse!

TRIP Density

Definition

$$\text{Density} = \frac{2|E|}{|V|(|V|-1)} \quad \text{(undirected)}$$

Example

- Order: $|V| = 216$
- Size: $|E| = 362$

$$\text{Density} = \frac{2 \times 362}{216 \times 215} = \frac{724}{46440} \approx 0.0156$$

Interpretation

- Density $\approx 0.016 \to$ very sparse network
- Only about 1.6% of all possible edges are present

Clustering

Local Clustering Coefficient of a node i measures the fraction of pairs of neighbors of i that are connected to each other.

$$C_i = \frac{2T_i}{d_i(d_i - 1)},$$

where d_i is the degree of node i, and T_i is the number of triangles involving node i.

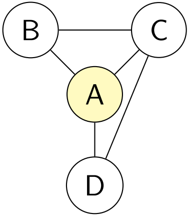
Global Clustering Coefficient is the average of local clustering coefficients.

Transitivity measures the overall probability that adjacent vertices of a vertex are connected.

$$T = \frac{3 \times \#\text{ triangles in G}}{\#\text{ connected triples of vertices}}.$$

Local Clustering Coefficient

Example Network



$$C(A) = \frac{2T_A}{d_A(d_A - 1)}$$

Step 1: Identify neighbors and degree

$N(A) = \{B, C, D\}, \quad d_A = 3$

Step 2: Count Triangles

- A-B–C $= 1$
- A-C–D $= 1$
- A-B–D $= 0$
- $\mathrm{T}_A = 2$

Step 3: Compute Clustering coefficient

$C_A = 2 * 2/(3 * (3 - 1)) \approx 0.67$

Global Clustering vs. Transitivity

Global Clustering Coefficient is the average of local clustering coefficients. *How clustered is a typical node's neighborhood?*

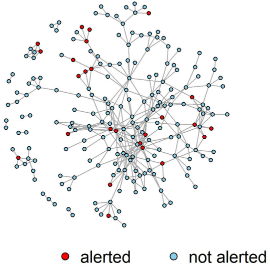
Transitivity is a global measure counting structures in the entire graph. *What is the probability that two neighbors of a node are connected?*

Key Differences:

- The global clustering is an *average of node-level quantities*.
- Transitivity is a *ratio of global counts*.
- Global clustering gives equal weight to all nodes, while transitivity gives more weight to high-degree nodes.

TRIP Network Characteristics

Order (Nodes)	216
Size (Edges)	362
Density	0.016
Components	10
Average Degree (SD)	3.35 (2.75)
Transitivity	**0.25**
Alerted	25(11.6 %)
Not Alerted	191(88.4%)



- Weak-to-moderate clustering (not random) sufficient to sustain local outbreaks within subgroups.
- HIV transmission likely occurs in partially overlapping risk clusters, not fully closed communities.

Centrality Measures

- **Closeness** measures how close a node is to all other nodes

$$C_C(i) = \frac{1}{\sum_{i \in V} dist(i,j)},$$

where $dist(i,j)$ is the shortest path between nodes i and j.

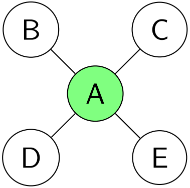
- **Betweenness:** measures how often a node lies on shortest paths between other nodes.

$$C_B(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(i)$ is the number of those paths passing through v.

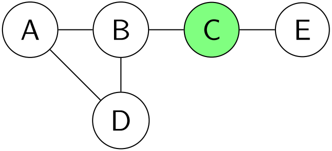
Closeness vs Betweenness Centrality

High Closeness Centrality



- Node A is **close** to all others.
- Short paths to every node.
- High efficiency of *reachability*.

High Betweenness Centrality



- Node C lies on many shortest paths.
- Acts as a **bridge** between groups.
- Controls flow between sub-networks.

TRIP Centrality Analysis

Skov B, Buchanan AL, Katenka NV, Hoque NT, Friedman SR, Nikolopoulos GK. *Evaluation of the Relationship Between Network Centrality and Individual Sociodemographics and Behaviors Among People Who Inject Drugs*. Substance Use & Misuse. 2026;61(3):357–365.

- Examines how sociodemographic characteristics and behaviors are associated with network centrality among PWID.
- Uses logistic regression to evaluate associations between high centrality and individual characteristics/behaviors.

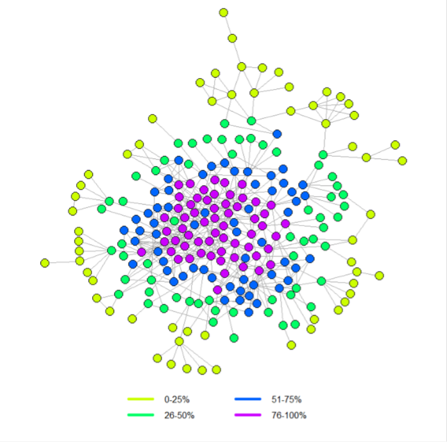
Key Results

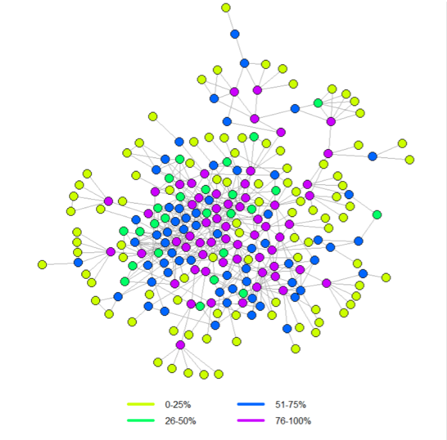
1. Individuals injecting at least once daily were more likely to have:
   - High closeness ($OR = 3.36$) and high betweenness ($OR = 2.22$)
2. Individuals engaging in condomless sex were less likely to have:
   - High closeness ($OR = 0.18$) or high eigenvector centrality ($OR = 0.19$)

TRIP Centrality Analysis

TRIP network LCC with node color based on centrality quartiles

High Closeness Centrality **High Betweenness Centrality**





Assortativity in Networks

Degree Assortativity: measures the tendency of nodes to connect to other nodes with similar degree. It is defined as the *Pearson correlation* between the degrees of pairs of connected nodes:

$$r = \frac{\sum_{(i,j)\in E}(d_i - \bar{d})(d_j - \bar{d})}{\sum_{(i,j)\in E}(d_i - \bar{d})^2},$$

where d_i and d_j are the degrees of the endpoints of an edge, and $\bar{d}$ is the mean degree over edges.

TRIP **degree assortativity = 0.25**

- Moderate positive tendency for nodes to connect to others with a similar number of contacts.

Assortativity in Networks

Nominal Assortativity: measures the tendency of nodes to connect to others with the same categorical attribute (e.g., gender, type, group membership). It is defined as

$$r = \frac{\mathrm{Tr}(e) - \|e^2\|}{1 - \|e^2\|},$$

where e_{ij} is the fraction of edges connecting nodes of type i to type j, $\mathrm{Tr}(e)$ is the trace of e, and $\|e^2\|$ is the sum of squared row/column sums.

Interpretation:

+ **Positive r:** nodes in the same category tend to connect.
- **Negative r:** nodes in different categories tend to connect.

Graph Partitioning

- Refers to the segmentation of a set of elements into natural **cohesive** subsets.
- Concerned with the case where we *do not know node attributes*, but only the connectivity structure.

A **cohesive subset** of vertices refers to a subset where vertices are:

- well-connected among themselves, and
- relatively well-separated from the rest of the graph.

This is commonly referred to as *community detection* and is closely related to the general clustering problem.

Graph Partitioning Algorithms

Typically, seek a partition

$$\mathcal{C} = \{C_1, \ldots, C_K\}$$

of the vertex set V of a graph $G = (V, E)$ such that:

- Edges connecting nodes in different groups are relatively small.

$$E(C_k, C_{k'}) = \{(i,j) \in E : i \in C_k,\ j \in C_{k'}\}$$

- Edges $E(C_k) = E(C_k, C_k)$ connecting nodes within the same group are relatively large.

Methods of Graph Partitioning

A. Hierarchical clustering,

B. Modularity-based methods,

C. Spectral methods; and

D. Stochastic block models.

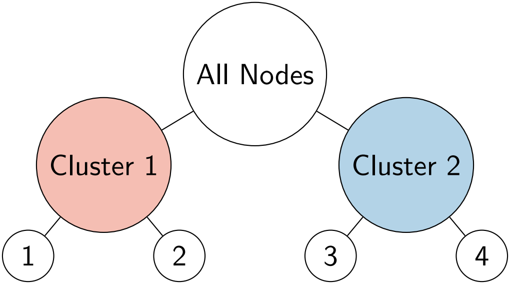
Hierarchical Community Detection in Networks

Networks often contain communities within communities and hierarchical methods uncover **multi-level (nested)** structure.

Approach

- Start with full network as one group
- Recursively split into smaller communities (top-down), or
- Merge small communities into larger ones (bottom-up)

Example of hierarchical structure



Modularity-Based Clustering

Goal: Find communities where edges are denser than expected under a random network.

Key idea: Compare observed edges to a null model.

$$Q = \frac{1}{2|E|} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2|E|} \right) \delta(c_i, c_j)$$

- A_{ij}: observed adjacency
- d_i: degree of node i
- $|E|$: size/total number of edges
- $\delta(c_i, c_j)$: 1 if same community

Intuition Behind Modularity

Goal: By maximizing Q, partition the network to maximize within-community edges and minimize between-community edges.

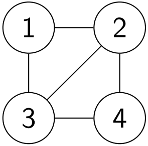
$$Q = \frac{1}{2|E|} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2|E|} \right) \delta(c_i, c_j)$$

+ **Positive Q:** more within-community edges than expected
- **Negative Q:** worse than random grouping
- Communities = groups with unexpectedly many internal connections
- Compared against a "degree-preserving random graph"

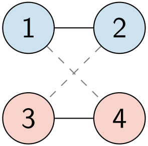
Modularity Optimization: Recursive Partitioning

Greedy idea (e.g., Louvain method)

Original Network



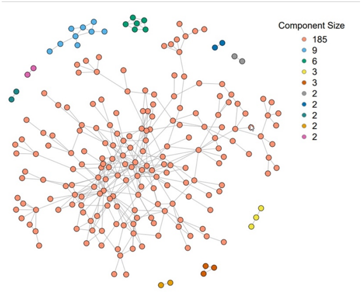
After Partitioning

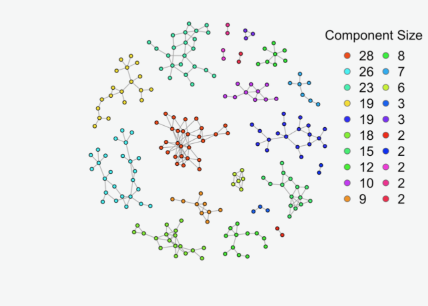


1. Start with each node in its own community
2. Move nodes to neighboring communities if Q increases
3. Aggregate communities into "super-nodes"
4. Repeat until no improvement in Q

- High-modularity partition of the network
- Communities maximize internal density
- Cross edges are minimized

Connected Components and Communities





- TRIP has **10** components.
- Using modularity-based community detection divided the network into **20** components.

Limitations of Modularity-Based Clustering

- **Resolution limit:** small communities may be missed
- **Multiple optima:** different partitions can have similar Q
- **Non-uniqueness:** results depend on algorithm initialization
- **Not probabilistic:** lacks explicit generative model

Puleo, J.; Buchanan, A.; Katenka, N.; Halloran, M.E.; Friedman, S.R.; Nikolopoulos, G. Assessing Spillover Effects of Medications for Opioid Use Disorder on HIV Risk Behaviors among a Network of People Who Inject Drugs. Stats 2024, 7, 549-575.

Note: The magnitudes of estimated effects were sensitive to the community detection method. Careful consideration should be paid to the significance of community structure in network studies evaluating spillover.

Spectral Clustering Intuition (Graph Laplacian)

Key idea: Find eigenvalues of $L = D - A$:

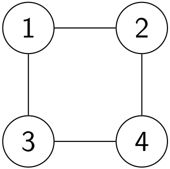
$$Lv = \lambda v$$

Interpretation:

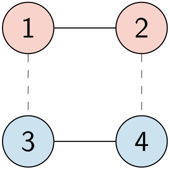
- Smallest eigenvalue is always 0
- Number of near-zero eigenvalues $\rightarrow$ number of clusters
- Second smallest eigenvalue (**Fiedler value**) measures connectivity
- If Fiedler value is small $\rightarrow$ weakly connected groups
- Eigenvectors of L embed nodes into low-dimensional space
- Clustering is done in this spectral space

Spectral Clustering: Two-Cluster Split

Original Graph



Spectral Clusters



- Eigenvalues of L: $0, 2, 2, 4$
- Fiedler value: $\lambda_2 = 2$
- **Fiedler vector (associated with λ_2):** $v_2 \approx (1,\ 1,\ -1,\ -1)$

Clustering rule:

- Positive entries $\to$ Cluster 1: nodes $\{1, 2\}$
- Negative entries $\to$ Cluster 2: nodes $\{3, 4\}$

Key intuition: The sign structure of the second eigenvector reveals the natural graph partition.

Network Graph Models

Generally, takes the form

$$\{P_\theta(G),\ G \in \mathcal{G} : \theta \in \Theta\}.$$

- Used to study mechanisms that produce observed network characteristics.
- Test the significance of observed characteristics.
- Assess associations between network structure and node/edge attributes,
- Generate synthetic networks (for modeling processes/flows).

Two Types of Network Models

- Mathematical/Probabilistic Models
- Statistical Models

Mathematical Network Models

- **Erdős–Rényi Model (or Random Graph):** Independent edges with probability p; used to study connectivity and phase transitions.
- **Configuration Model (or Generalized Graph):** Generates graphs with a fixed degree sequence; captures degree heterogeneity.
- **Preferential Attachment Model:** New nodes attach to high-degree nodes; produces power-law (scale-free) networks.
- **Small-World Model (Watts–Strogatz):** Combines local clustering with short path lengths.
- **Percolation Models**, **Graphon Models**, and **Branching Processes** .

Mathematical Network Models

Mathematical/Probabilistic Models focus on constructions that enable mathematical analysis of structural properties.

A **Erdős–Rényi Random Graph** is a graph with N_v nodes in which:

- each pair of nodes is linked independently with probability p,
- N_v nodes are connected by N_e edges placed uniformly at random.

Random Graphs are often sparse and exhibit low clustering with the degree following Poisson distribution.

Real Networks are usually not Random Graphs

- Poisson distribution underestimates the number of high-degree nodes.
- Degree spread is wider than expected in a random network.
- Transitivity is higher than would be expected in a random network.

Characteristic	TRIP	Erdős–Rényi Random Graph
Order (Nodes)	216	216
Size (Edges)	352	352
Density	0.016	0.016
Components	10	8
Average Degree (SD)	3.35 (**2.75**)	3.25 (**1.75**)
Transitivity	**0.25**	**0.016**
Assortativity	**0.25**	**0.085**

Random graph models are used to assess the significance of observed network characteristics (serving as null models).

Statistical Network Models: Motivation

Why Statistical Network Models?

- Network data violates independence assumptions.
- Observations are dependent and collected at different levels (nodes/edges/communities).
- Need probabilistic/inferential models for entire graphs.

Quick Decision Guide:

- Local structural effects on edge formation $\rightarrow$ Exponential Random Graph Models (ERGMs)
- Community structure or clustering $\rightarrow$ Stochastic Block Models (SBMs)
- Continuous similarity in hidden space $\rightarrow$ Latent Space Models

Statistical Network Models

Statistical Network Models focus on statistical estimation, inference, and interpretation.

- **ERGMs:** Model network structure using statistics such as edges, triangles, and homophily, allowing dependence among edges.
- **SBMs:** Assume nodes belong to latent groups, with edge probabilities driven by group membership.
- **Latent Space Models:** Place nodes in a latent geometric space where edge probabilities depend on distance, capturing similarity and transitivity.

Exponential Random Graph Models

Primary Goal: Identify features of the observed network that distinguish it from other random networks with the same nodes (and often similar density).

- ERGMs model how node and edge attributes shape edge formation and network structure.
- Similar to logistic regression, but without assuming independent edges.
- The observed network is treated as one realization from a multivariate distribution over *population of networks*.

Major Concern: Model degeneracy and computationally intensive MCMC-based fitting.

Exponential Random Graph Models

Let $G = (V, E)$ be a random graph, $\mathbf{Y} = [Y_{ij}]$ the corresponding random adjacency matrix (matrix of 1s and 0s).
Exponential Random Graph Model (ERGM) for $\mathbf{Y}$ is defined as

$$\Pr(\mathbf{Y} = \mathbf{y}) = \frac{\exp\left(\boldsymbol{\theta}^\top g(\mathbf{y})\right)}{\kappa(\boldsymbol{\theta})},$$

- $\mathbf{y}$ is an observed network,
- $g(\mathbf{y})$ is a vector of network statistics (structural network/node/edge characteristics or their summaries),
- $\boldsymbol{\theta}$ is the vector of coefficients, and
- $\kappa(\boldsymbol{\theta})$ is a normalization constant, generally unknown.

ERGMs: Alternative Definition

An ERGM can also be defined conditionally as

$$\text{logit}\left[\Pr\left(Y_{ij} = 1 \mid, \mathbf{Y}^c\right)\right] = \boldsymbol{\theta}^\top \Delta g(\mathbf{y})_{ij},$$

where

- Y_{ij} is an actor pair in $\mathbf{Y}$,
- $\mathbf{Y}^c$ denotes the rest of the network (without edge (i, j)),
- $\Delta g(\mathbf{y})_{ij}$ is the change in $g(\mathbf{y})$ when Y_{ij} is switched on.

Edges are generally **not independent**. Dependence is introduced through the chosen network statistics $g(\mathbf{y})$.

- **Edges**: baseline connectivity/density
- **Triangles**: transitivity/clustering
- **k-stars**: degree heterogeneity
- **Homophily**: similarity-based ties

Bernoulli Random Graph Assumptions

For any given pair of nodes, the presence of an edge is **independent** of the possible edges between any other node pairs:

$$Y_{ij} \perp Y_{i'j'}, \qquad \text{for } \{i', j'\} \neq \{i, j\}.$$

Homogeneity across mode pairs:

$$\theta_{ij} \equiv \theta, \qquad \text{for all } \{i, j\},$$

- The model contains only the total number of edges as a covariate (the simplest model).
- The coefficient θ (similar to the intercept β_0 in logistic regression) corresponds to the **log-odds** of a present edge without any other node/edge characteristics and can be translated into edge probability (or density):

$$p(\text{edge}) = \text{density} = \frac{e^{\theta_{\text{edges}}}}{1 + e^{\theta_{\text{edges}}}}.$$

Markov Random Graph Models

Frank and Strauss introduced the notion of Markov dependence for network graph models (Markov graphs):

- the presence or absence of $\{i, j\}$ in the graph depends upon that of $\{i, k\}$, for a given $k \neq j$.

Under homogeneity, Frank and Strauss showed G is a Markov graph if and only if

$$P_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y}) = \frac{1}{\kappa(\boldsymbol{\theta})} \exp\left(\sum_{k=1}^{N_v - 1} \theta_k S_k(\mathbf{y}) + \tau T(\mathbf{y}) \right),$$

where

- $S_1(\mathbf{y}) = N_e$,
- $S_k(\mathbf{y})$ is the number of k-stars, for $2 \leq k \leq N_v - 1$,
- $T(\mathbf{y})$ is the number of triangles.

Attribute-Based Models

Attribute-based effects can also be incorporated into ERGMs:

$$P_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y}) = \frac{1}{\kappa(\boldsymbol{\theta})} \exp\left(\boldsymbol{\theta}^\top g(\mathbf{y}, \mathbf{x})\right).$$

Note: Main effects and second-order effects may be incorporated into a model within `ergm` using:

- `nodecov` for numerical attributes,
- `nodefactor` for categorical variables,
- `nodematch` for homophily effects.

We will start with the simplest model (with only the edges term) and one covariate term.

ERGMs and HIV Risk Behaviors Among PWID

Ryan V., Lee T., Piovani D., Katenka N., Friedman S.R., Bonovas S., Buchanan A., Nikolopoulos G. *Attributes Associated with HIV Risk Behaviors in a Network-Based Study of People Who Inject Drugs.*

- Uses ERGMs to evaluate associations between individual-level characteristics and the likelihood of forming HIV risk ties among PWID.

Key ERGM Findings

- Positive degree assortativity: participants tended to connect with others having similar numbers of connections.
- Evidence of homophily: participants were more likely to form ties with individuals who were similar in: sex, nationality, and housing/living conditions.
- Findings may inform the design of future HIV prevention and intervention strategies.

Textbooks

- Kolaczyk, E. D. (2009). *Statistical Analysis of Network Data: Methods and Models*.
- Kolaczyk, E. D., & Csárdi, G. (2020). *Statistical Analysis of Network Data with R* (2nd ed.).
- Barabási, A.-L. (2016). *Network Science*.
- Durrett, R. (2007). *Random Graph Dynamics*.
- Hamilton, W. L. (2020). *Graph Representation Learning*.
- Wasserman, S., & Faust, K. (1994). *Social Network Analysis: Methods and Applications*.

Web Resources

- **General Network Science**
  - *Network Science (Barabási)* – free online textbook covering fundamental concepts such as degree distributions, clustering, and community structure.
- **Python Ecosystem**
  - `NetworkX` – core library for graph algorithms and network analysis (centrality, clustering, shortest paths).
  - `PyTorch Geometric` – library for graph neural networks and deep learning on graphs.
- **R Ecosystem**
  - `igraph` – efficient library for large-scale network analysis.
  - `statnet` – suite of packages for statistical network modeling, including ERGMs.

Thank you! Questions?

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